Uber Ride-Hailing Operations

Business Analysis Case Study — Self-Initiated Portfolio Project

Scope: Simulated ride-hailing operations analysis using synthetic data. This project was not performed for or on behalf of Uber.

1. Business Problem & Objectives

Analyse a ride-hailing operating model to identify opportunities to improve trip fulfilment, customer experience and operational visibility. The project focuses on ride requests, driver matching, acceptance, cancellations, completed trips and revenue.

Success measures: improve visibility of completion/cancellation performance, identify operational hotspots, and provide a repeatable KPI framework for decision-making.

2. Stakeholders

Stakeholder	Interest / Need
Operations Manager	Trip fulfilment, cancellations, driver supply, exceptions
Drivers	Acceptance opportunities, utilisation, demand patterns
Customer Support	Cancellation and service-quality patterns
Business Leadership	Revenue, growth, operational KPIs
Customers	Reliable pickup and trip completion

3. Requirements

- Dashboard shall provide total trip volume, completion rate, cancellation rate and revenue.
- Users shall be able to analyse KPIs by city and month.
- Users shall be able to identify cancellation hotspots.
- Dashboard shall support operational investigation of driver acceptance and trip completion.
- KPIs shall use consistent definitions and be suitable for management reporting.
- Business recommendations shall be linked to observed performance patterns.

4. KPI Definitions

KPI	Definition	Business Use
Completion Rate	Completed / requested trips	Service fulfilment
Cancellation Rate	Cancelled / requested trips	Customer & operational risk
Driver Acceptance Rate	Accepted / requested trips	Driver supply / matching
Revenue	Completed-trip fare revenue	Commercial performance
Average Trip Value	Revenue / completed trips	Unit economics
Trip Volume	Count of requests	Demand
Avg Trip Duration	Mean duration of completed trips	Operational efficiency

5. User Stories & Acceptance Criteria

US01 — As an Operations Manager, I want to see completion and cancellation rates by month so that I can identify deteriorating service performance.

Acceptance criteria: Given trip data exists, when a month is selected, then the dashboard shows requested, completed and cancelled trips plus the calculated rates.

US02 — As an Operations Manager, I want to compare cancellation rates by city so that I can prioritise investigation.

Acceptance criteria: Given multiple cities exist, when the city view is opened, then cancellation rate and trip volume are shown for each city.

US03 — As a Business Manager, I want to track revenue and average trip value so that I can monitor commercial performance.

Acceptance criteria: Revenue and average trip value must update correctly when date or city filters are applied.

US04 — As an Operations Analyst, I want to monitor driver acceptance so that I can identify potential supply or matching issues.

Acceptance criteria: The report must calculate accepted requests as a percentage of total requests and allow trend analysis.

US05 — As a Customer Support Lead, I want to identify cancellation hotspots so that recurring service issues can be investigated.

Acceptance criteria: The report must allow cancellation analysis by city and reporting period.

US06 — As a Business User, I want consistent KPI definitions so that reports can be compared over time.

Acceptance criteria: Each KPI must have a documented calculation definition and use the same logic across report pages.

US07 — As an Operations Manager, I want an executive summary so that I can quickly identify performance changes.

Acceptance criteria: The overview page must surface key KPIs, trends and the most significant operational exceptions.

US08 — As a Data Analyst, I want the data transformed consistently so that Power BI calculations are reliable.

Acceptance criteria: Power Query transformations must standardise fields, types and derived reporting attributes.

6. MoSCoW Prioritisation

Priority	Requirements
Must	Core KPI dashboard; consistent KPI definitions; monthly/city filtering; cancellation and completion analysis.
Should	Driver acceptance analysis; exception-focused operational view; documented data model.
Could	Additional segmentation, advanced drill-through, automated alerts and richer demand forecasting.
Won't (for this prototype)	Production integration with live Uber systems, real-time driver dispatch or predictive optimisation.

7. Process Analysis

The As-Is process highlights potential friction around driver matching, acceptance and cancellation. The To-Be process introduces demand signals, smarter matching logic and KPI-driven exception monitoring. The diagrams are intentionally simplified to communicate business analysis thinking rather than claim Uber's actual internal process.

8. Power BI Prototype

The prototype uses synthetic trip-level data and demonstrates KPI cards, monthly trends, city comparisons and cancellation analysis. The dashboard is designed as an operational decision-support tool, not merely a visual report.

Suggested Power BI pages: Executive Overview; Operational KPIs; Cancellation Analysis; City / Demand Analysis.

9. Recommendations

- Investigate high-cancellation cities and periods using driver acceptance and demand indicators.
- Use a standard operational KPI layer so managers can compare performance consistently.
- Monitor driver acceptance alongside cancellation and completion rates to identify supply/matching issues.
- Use the dashboard for exception management: focus operational attention where KPIs deteriorate.
- Validate all findings against production-quality data before making real operational decisions.

10. Limitations & Assumptions

The dataset is synthetic and created for portfolio demonstration. It does not represent Uber's actual performance, internal processes, systems or confidential data. The case study demonstrates Business Analysis, process modelling, requirements definition and Power BI skills.